

1. Forman 10:17



Topic:- Data Science & AI

Data Science :- Data + Science

Data (Types of Data)

1) Excel data $\rightarrow$ stored in data base
(data generate from customer)

It is called as Structured data

Structured data = Supervised learning
= labeled data

2) Image data :- Unstructured data
= unsupervised learning = unlabeled data

3) Video data :- unstructured data
= unsupervised learning = unlabeled data

4) Audio Data :- unstructured data

5) Streaming data :- Un structured data

6) Sensor Data :- un structured data

7) XML/ PDF :- un structured data

Topic :- Data Science & AI

Topics for Today Class

data science architecture

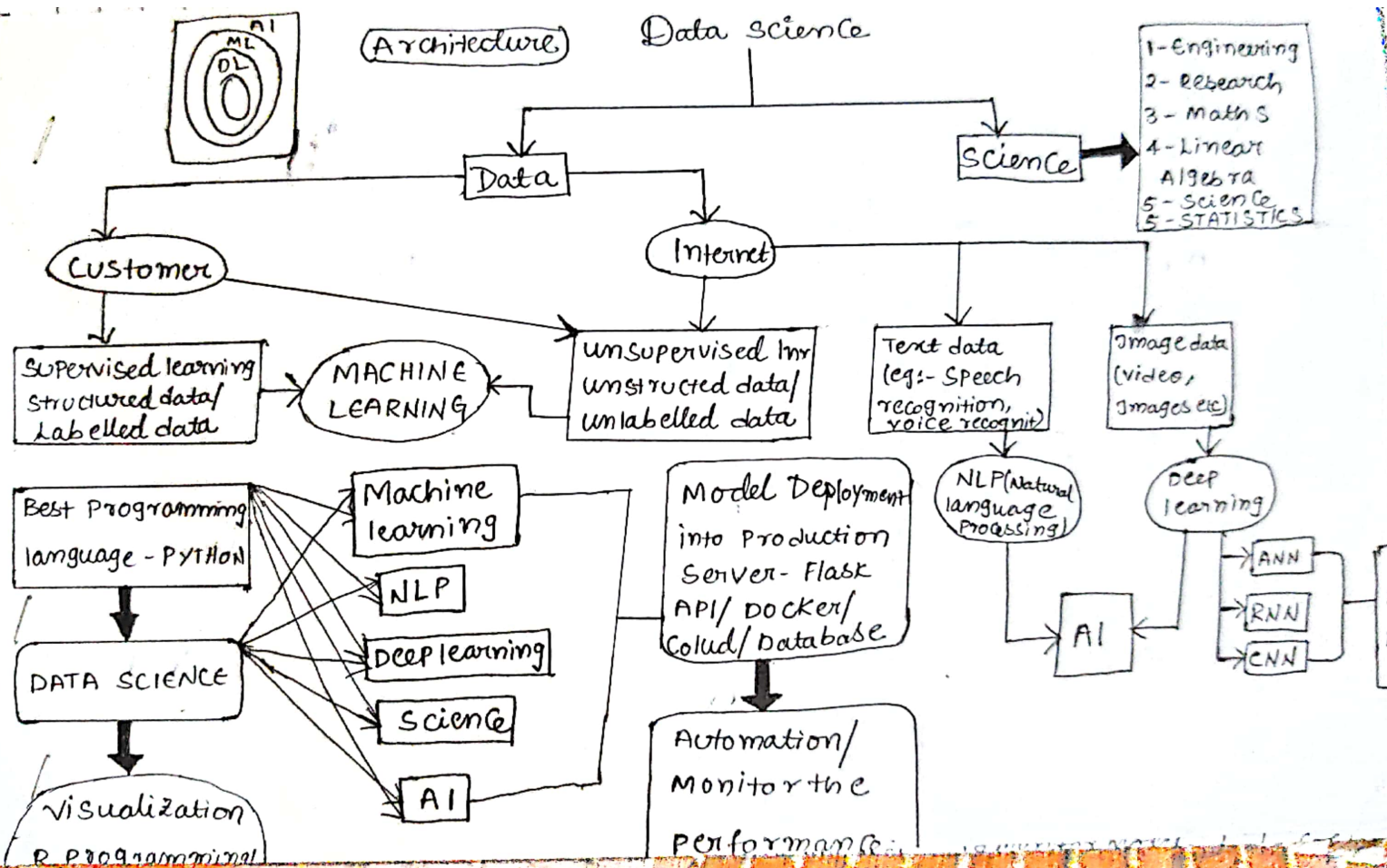
data science pipeline

data science $\rightarrow$ ml + nlp + dl + nn + ai +

Statistic $\rightarrow$ Python (99%) R, Spark,

Azure ML

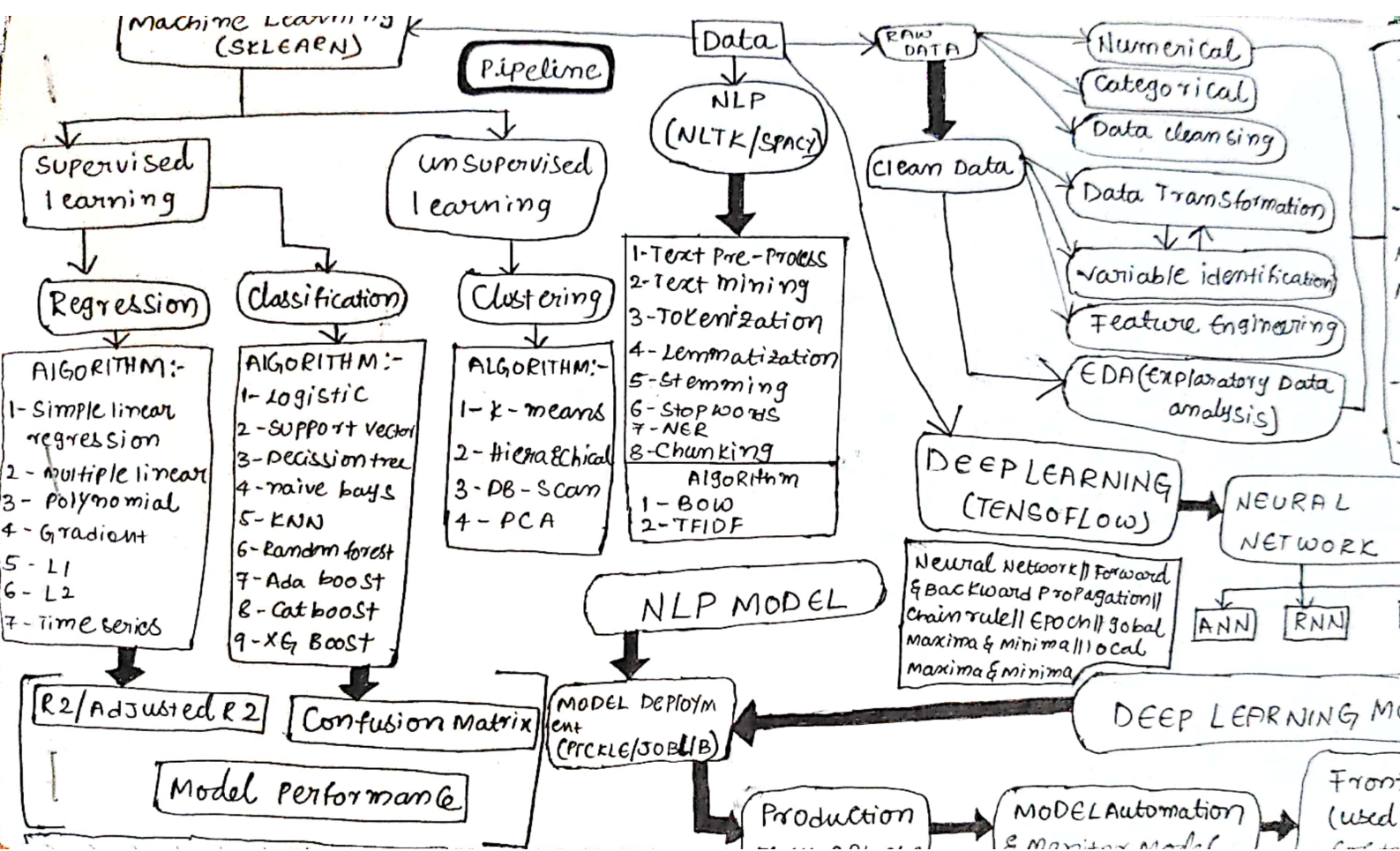
--- once we build the model visualize
the model tableau.

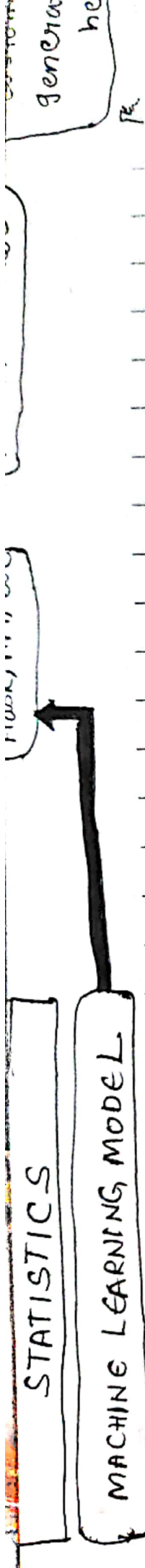


Learning points in the course:-

- Python Programming language
- Statistics
- Machine Learning
- Natural Language Processing
- Deep learning
- Computer Vision
- Deep Net
- Reinforcement learning (Open AI)
- GAN (Generative Adversarial Network)
- Time Series
- Data Analytics
- Business analytics
- Artificial Intelligence
- R Programming language
- Deploy model using Flask and HTML
- Azure machine Learning
- Apache Hadoop
- Apache Spark
- Data Bricks
- Tableau
- Structured Query Language (SQL)
- Advance Excel
- Capstone Projects - BFSI
- Advanced Computer Vision Project
- 1 on 1 Resume Preparation
- Mock interview session
- Project discussion session
- Resume building after complete modules

Tableau/Python
EDA





Sklearn - framework for
Machine learning

nltk :- framework for text data

tensorflow :- framework for
deep learning.

Server → it is machine which
is on for 24/7

iPhone amazon

amazon box (Package)

module (iPhone)

phone (function)

Package → collection of module

module → collection of function

Topic :- Data Science & AI

Python index begins from 0

forward index $\rightarrow 0$

backward index $\rightarrow -1$

F.W	0	1	2	3	4	5	6	7	8	9	10	11
	H	E	L	L	O		P	J	T	H	O	N
	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1 B.W

Artificial Intelligence (AI)

The term AI introduced by John McCarthy in the Year of 1956

AI is a technique of getting machines to work and behave like humans.

AI is the combination of science and engineering of making intelligent machines using algorithms.

Road Map TO Data Scientist :-

Python → eda → sql → statistic
→ machine learning (regress,
classificat, clustering) → flask &
html → nlp → deep learning
→ neural network → time
series forecasting → spark &
hadoop - (azur ml cloud) → tableau
→ R programming.

Demo class - 04, 05, 06, 07

05:- 12-07-2021

(08) → last Demo
15-07-2022
Friday

Tuesday

Wednesday

06:- 13-07-2022

11-07-2022

Monday

Topic :- Data Science and AI

import sys
sys.version } → TO check anaconda
version

My updated version

3.9.12 [APR 4, 2022]

Rules to Define python Identifier

1) Identifier should not start with number/digit

2) no special characters is ^{not} allowed

3) Identifier is case sensitivity/sensitive

4) Key words can't be an identifier

Identifier = variable = object

5) No length limit in Python identifier

6) '-' underscore is allowed in Python identifier


```
import keyword  
keyword.kwlist
```

To check
Key words
of
Python

Python Reserved words / Key words

→ True, false, None → Represent Boolean data types

→ and, or, not, is ⇒ Represent the operators

→ if, else, elif ⇒ Represent the Statement

⇒ while, for, break, continue, return, in, yield ⇒ Represent the loop concept

⇒ try, except, finally, raise, assert
⇒ Represents for functionality.

⇒ import, from, as, class, def, pass, global, nonlocal, lambda, del, with ⇒ Represent the class

```
import pandas as pd
```

```
df = pd.DataFrame(keyword.kwlist)
```

```
df
```


Demo class :- 04, 05, 06, 07

(08) → labt Demo
15-07-2022
Friday

Data Science & AI

Pandas --- dataframe

python index beginning from --- 0

Plot the graph using 1 variable

- Statics (univariate analysis)

Plot the graph using 2 variables

- Statics (bivariate analysis)

Graph using 10 variables [HEATMAP]

Column name = variable = attribute
= features

python Data type

int || float || String || complex || bool

Int → the value without decimal

a = 10

binary format

float datatypes → value with decimal
is Petrol, Price, Salary, 109.110

- binary, octal do not work with float datatype

- e or E letter is allowed in float

Complex data types :-

format of complex data type
(a+bj)

a - Real part

b - Imaginary part

j - is the compulsory value and there is no other value accepted in complex type

$$j^2 = -1$$

- Imaginary part cannot be binary, octal, it allows only integer

→ Real part can allow binary and octal.

$$a1 = 10 + 20j$$

$$a1.real = 10.0 \quad \} \rightarrow \text{output}$$

$$a1.imag = 20.0 \quad \}$$

Bool data types :-

True / False →

False value → 0 (Internally)

True value → 1

True/False = 1/0 = Zero division error

Str data types :-
STRING

→ enclosed in 'single' or "double" quotes.

→ single & double quotes will not work for multiline comments

→ only triple quotes will work for multiline comments.

Type Casting or Type Conversion :-

int() -- float() -- Complex() -- bool()
-- String()

Type casting is trying to ~~one~~ convert one data type to other data type

int(10.123) → float to int

int(10+20j) → we can't convert from

int(True) → Bool to int

int(False) → Bool to int

int('11') → String to int

$\text{float}(10) \rightarrow \text{int to float}$

$\text{float}(10+20j) \rightarrow \text{Cannot Convert Complex to float}$

$\text{float}(\text{False}) \rightarrow \text{Bool to float}$

$\text{float}('11') \rightarrow \text{String to float}$

1 $\text{complex}(10) \rightarrow \text{int to complex}$

2 $\text{complex}(10, 20) \rightarrow \text{int to complex}$

3 $\text{complex}(10.5) \rightarrow \text{float to complex}$

4 $\text{complex}(\text{True}) \rightarrow \text{bool to complex}$

5 $\text{complex}(\text{False}) \rightarrow \text{bool to complex}$

6 $\text{complex}('10') \rightarrow \text{String to complex}$

o/p

1) $10+0j$ 2) $10+20j$ 3) $10.5+0j$

4) $1+0j$ 5) $-0j$ 6) $10+0j$

o/p

$\text{bool}(0) \rightarrow \text{int to bool} \rightarrow \text{False}$

$\text{bool}(10) \rightarrow \text{int to bool} \rightarrow \text{True}$

$\text{bool}(0.0) \rightarrow \text{float to bool} \rightarrow \text{False}$

$\text{bool}(0.01) \rightarrow \text{float to bool} \rightarrow \text{True}$

$\text{bool}(10+20j) \rightarrow \text{Complex to bool} \rightarrow$

$\text{bool}(0+1j) \rightarrow \text{Complex to bool} \rightarrow$

$\text{bool}("") \rightarrow \text{String to bool} \rightarrow \text{False}$

$\text{bool}(" ") \rightarrow \text{String to bool} \rightarrow \text{True}$

o/p

`Str(10) → int to String → '10'`
`Str(10.50) → Float to String → '10.5'`
`Str(True) → bool to String →`
`Str(10+20j) → Complex to String →`

- 1) Convert other data type to int except complex
- 2) Convert other data type to float except complex
- 3) Convert other data types to complex
- 4) Convert other data types to bool
- 5) Convert other data types to string.

Data Structures :-

* Collection of data types is called as Data Structures.

Types of Data Structures in Python

- 1) list
- 2) tuple
- 3) set
- 4) frozen set
- 5) dict (dictionary)
- 6) range

Class - 01

* Append object / value to the end of the list

18-07-2022

Monday

D.S
List of

Topic :-

Data Science & AI

10, 20, 30, 10, 8.0, None

LIST (Data Structure) :-

slicing operator
[:]

Printing the

→ LIST denotes as a []
Square bracket

→ Duplicates are allowed in the list

[10, 20, 10, 20, 50]

→ Multiple data types are allowed

→ Indexing is allowed

→ append is allowed

→ remove is allowed

→ List is mutable / changeable

Mutable means once you assigned the data type in data structure can be changed as our wish

19/07/2022 - Tuesday

TUPLE (Data Structure) :-

- Tuple denotes as a () open brackets
- duplicates are allowed
- Multidata types are allowed
- It is immutable / unchangeable
- append is not allowed
- remove is not allowed
- add is not allowed
- indexing is allowed
- slicing is allowed

Eg :- Adhar card, Pan card.

20/07/2022 - Wednesday

Range (Data Structure)

- In Range we can pass only 3-arguments
- float argument is not allowed in range (float datatype).

append $\rightarrow$ add the element in last
add $\rightarrow$ Add an element

SET (Data Structure)

$\rightarrow$ It Denotes as $\{ \}$ (Flower brackets)

$\rightarrow$ Set dose not SUPPORT indexing or slicing

$\rightarrow$ multiple Data types is allowed

$\rightarrow$ Insertation order are not preserved, order & duplicates are not allowed

$\rightarrow$ append is not allowed

$\rightarrow$ Set is mutable / $\rightarrow$ mutable

$\rightarrow$ add & remove method is use for set is allowed.

FROZEN SET (Data Structure)

$\rightarrow$ once variable is freezed

we cannot add or remove

and we can not unfreezed.

S H F D B

Date: _____
Page: _____

Dictionary (Dict) (Data Structure)

→ It denotes as $\{ \}$ (flower brackets)

→

25-07-2022 Monday

Numpy Session (Numpy)

Numpy is the package which is used for multidimension array
matplotlib $\rightarrow$ visualization

import numpy as np

Seaborn $\rightarrow$ used for advanced visualization
Creating Arrays

```
my_list = [0, 1, 2, 3, 4, 5]
```

```
my_list
```

```
o/p = [0, 1, 2, 3, 4, 5]
```

```
arr = np.array(my_list) arr
```

```
o/p :- array([0, 1, 2, 3, 4, 5])
```

```
np.arange(0, 10)
```

arange :- function

```
o/p :- array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
np.arange(0, 10, 5)
```

```
o/p :- array([0, 5])
```

```
np.zeros((4, 4))
```

$\rightarrow$ matrix

zeros & ones
are Matrix

```
o/p array([[0, 0, 0, 0],  
           [0, 0, 0, 0],  
           [0, 0, 0, 0],  
           [0, 0, 0, 0]])
```

```
import matplotlib.pyplot as plt
```

```
np.zeros((5, 4))
```

↑ rows
↑ columns

```
np.ones((2, 4))
```

```
array([[1, 1, 1, 1],  
       [1, 1, 1, 1]])
```

Colour Change
plt.plot
(Salary[0], c='red')

26-07-2022 Tuesday

MATRICES/ NUMPY - - -

X-ticks → X-axis, Y-ticks → Y-axis

→ Matrix is the tabular representation of the data

→ lot of data are stored in table format that is why matrices is very important

→

28th JULY-2022

Pandas --> Data frame

Import Pandas as Pd

Pandas is a software data manipulation and analysis tool used for data frame.

EXPLORATORY DATA ANALYSIS

Row data vs Column data

Column = attribute = feature = Variable

→ Every time build the model or build the visualization with Clean data.

- distplot → distribution plot

- len - total number of row

- Shape - rows & columns

- info -- information of the dataset // by default understands Categorical

(dtype - object type)

- columns -- display column names in the data set

- head -- it will give top 5 rows

- tail - display last 5 rows

- describe - it will display descriptive statistics information of the data set

- drop - remove from dataframe

Stats -- descriptive statistic vs

inferential stats describe

always gives the numerical value

outlier :- An outlier is a data point